

# An Exact Decomposition of the Divisor Summatory Function and Fixed- $j$ Asymptotics

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June 26, 2026

ABSTRACT. In this paper, we study a decomposition of the divisor summatory function  $D(x)$ . We establish the exact identity

$$D(x) = 1 + \sum_{i \geq 1} (i+1)\pi(x^{1/i}) + \sum_{j \geq 2} (-1)^j (j-1)T_j(x),$$

which decomposes  $D(x)$  via inclusion–exclusion according to the number of distinct prime factors of  $n$ . The three terms correspond respectively to  $n = 1$ , prime powers, and integers with  $\omega(n) \geq 2$ .

We then establish an asymptotic formula with a stated error term for  $T_j(x)$  with fixed  $j \geq 2$ . We consider the two-variable Dirichlet series  $F(s, y)$  and obtain the factorization  $F(s, y) = \zeta(s)^{2+2y} H(s, y)$ , where  $H$  is holomorphic near  $s = 1$ . Applying the Selberg–Delange method—a truncated Perron formula, contour deformation, and local Hankel analysis—we prove that

$$T_j(x) \sim \frac{2^j}{j!} x \log x (\log \log x)^j \quad \text{as } x \rightarrow \infty.$$

These estimates describe the individual fixed- $j$  components of the decomposition, rather than the full alternating sum.

## 1 Introduction

### 1.1 The Dirichlet divisor problem

Let  $d(n)$  denote the number of divisors of a positive integer  $n$ , and let

$$D(x) := \sum_{n \leq x} d(n) \tag{1.1}$$

be its summatory function. Dirichlet [2] proved in 1849 that

$$D(x) = x \log x + (2\gamma - 1)x + \Delta(x), \tag{1.2}$$

where  $\gamma$  is the Euler–Mascheroni constant and  $\Delta(x) = O(x^{1/2})$ . The Dirichlet divisor problem asks for the infimum  $\theta$  of all real numbers such that  $\Delta(x) = O(x^{\theta+\varepsilon})$  for every  $\varepsilon > 0$ .

Hardy [4] proved that  $\theta \geq 1/4$ , and it is conjectured that  $\theta = 1/4$ . In the other direction, Voronoï [8] proved the classical estimate

$$\Delta(x) = O(x^{1/3} \log x), \quad (1.3)$$

which implies  $\theta \leq 1/3$ . In 2003, Huxley [5] proved the bound

$$\theta \leq 131/416. \quad (1.4)$$

In this paper, rather than pursuing further improvements of  $\theta$ , we explore an alternative approach: an exact combinatorial decomposition of  $D(x)$  via inclusion–exclusion on the number of distinct prime factors of  $n$ .

## 1.2 A new decomposition

For a positive integer  $n$ , let  $\omega(n)$  denote the number of distinct prime factors of  $n$ . For each integer  $j \geq 2$ , define

$$T_j(x) := \sum_{n \leq x} d(n) \binom{\omega(n)}{j} \quad (1.5)$$

where  $\binom{\omega(n)}{j}$  denotes the usual binomial coefficient. The main identity of this paper is the following.

**Theorem 1.1.** For every real number  $x \geq 1$ ,

$$D(x) = 1 + \sum_{i \geq 1} (i+1) \pi(x^{1/i}) + \sum_{j \geq 2} (-1)^j (j-1) T_j(x) \quad (1.6)$$

where  $\pi(x)$  denotes the prime-counting function. Both sums have only finitely many nonzero terms for each fixed  $x$ : we have  $\pi(x^{1/i}) = 0$  for all sufficiently large  $i$ , and  $T_j(x) = 0$  once  $j$  exceeds the maximum possible value of  $\omega(n)$  among integers  $n \leq x$ .

The three terms on the right-hand side of (1.6) correspond, respectively, to  $n = 1$ , prime powers  $\omega(n) = 1$ , and integers with  $\omega(n) \geq 2$ . The proof of Theorem 1.1 is given in Section 2.

Dirichlet's estimate gives  $D(x) \sim x \log x$ . By the prime number theorem [1], the prime-power contribution satisfies

$$\sum_{i \geq 1} (i+1) \pi(x^{1/i}) \sim \frac{2x}{\log x}. \quad (1.7)$$

This is  $o(x \log x)$ . Thus the first two terms in the identity are negligible on the scale  $x \log x$ . The main contribution must therefore come from the alternating sum

$$\sum_{j \geq 2} (-1)^j (j-1) T_j(x) \quad (1.8)$$

after cancellation. As the asymptotics below show, the individual terms  $T_j(x)$  are large, and substantial cancellation is needed in the full alternating sum. The fixed- $j$  asymptotics proved below give information about individual terms, but they do not by themselves determine the full alternating sum.

**Example 1.2.** At  $x = 10000$ , direct computation gives

$$\begin{aligned} T_2(10000) &= 297,078 & T_3(10000) &= 140,640 \\ T_4(10000) &= 26,674 & T_5(10000) &= 1,216 \end{aligned}$$

and  $T_j(10000) = 0$  for  $j \geq 6$ . The prime-power contribution equals 2,711, and the identity (1.6) yields

$$1 + 2,711 + (T_2 - 2T_3 + 3T_4 - 4T_5) = 1 + 2,711 + 90,956 = 93,668 = D(10000),$$

illustrating Theorem 1.1. Moreover, the ratio of  $T_j(x)$  to the leading term in Theorem 1.3 converges to 1 only very slowly: at  $x = 10^9$ , this ratio is 0.43 for  $j = 2$  and 0.21 for  $j = 3$ . This slow convergence motivates the refined asymptotic expansion of  $T_j(x)$  developed in Section 6.

### 1.3 Main results

To analyze  $T_j(x)$ , we consider the two-variable Dirichlet series

$$F(s, y) := \sum_{n \geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s}.$$

Here and below, the definition of  $T_j(x)$  is extended to all  $j \geq 0$  by the same binomial-coefficient formula. The factor  $(1+y)^{\omega(n)}$  is chosen so that its expansion in  $y$  produces the coefficients  $\binom{\omega(n)}{j}$ ; in Section 4, the series  $F(s, y)$  will be used to study the corresponding generating partial sums. The series  $F(s, y)$  admits the factorization

$$F(s, y) = \zeta(s)^{2+2y} H(s, y), \tag{1.9}$$

where, for  $y$  in a small disk around 0, the function  $H(s, y)$  is holomorphic and uniformly bounded in a strip  $\operatorname{Re}(s) > 1 - \delta$ . This factorization isolates the singularity of  $F(s, y)$  at  $s = 1$ , and it is the starting point of the Selberg–Delange method [6, Chapter II.5].

Our main asymptotic result is the following.

**Theorem 1.3.** For each fixed integer  $j \geq 2$ , as  $x \rightarrow \infty$ ,

$$\begin{aligned} T_j(x) &= \frac{2^j}{j!} x \log x (\log \log x)^j \\ &\quad + \frac{2^{j-1}}{(j-1)!} (2(B_1 - 1) - P(2)) x \log x (\log \log x)^{j-1} \\ &\quad + O_j(x \log x (\log \log x)^{j-2}), \end{aligned} \tag{1.10}$$

where  $B_1$  is the Meissel–Mertens constant and  $P(2) = \sum_p p^{-2}$  is the value of the prime zeta function at 2. In particular,

$$T_j(x) \sim \frac{2^j}{j!} x \log x (\log \log x)^j. \tag{1.11}$$

The proof of Theorem 1.3 is given in Sections 4–6. The essential ingredients are the truncated Perron formula, contour deformation around  $s = 1$ , and a local Hankel analysis in the  $w$ -plane obtained via the substitution  $w = (s - 1) \log x$ .

Theorem 1.3 is a fixed- $j$  result. It describes the individual components  $T_j(x)$  in the decomposition, but it does not by itself determine the full alternating sum in (1.6). For further discussion of this limitation, see Section 7.

## 1.4 Outline of the paper

The paper is organized as follows. In Section 2, we prove the identity in Theorem 1.1 by partitioning  $D(x)$  according to  $\omega(n)$  and applying an inclusion–exclusion argument to the part with  $\omega(n) \geq 2$ . In Section 3, we introduce the two-variable Dirichlet series  $F(s, y)$  and establish its factorization  $F(s, y) = \zeta(s)^{2+2y}H(s, y)$ , together with the analytic properties of  $H(s, y)$  needed in the sequel.

Sections 4–6 are devoted to the proof of Theorem 1.3. In Section 4, we apply the truncated Perron formula for  $F(s, y)$  to represent the generating partial sum  $A(x, y) := \sum_{n \leq x} d(n)(1+y)^{\omega(n)}$  and deform the contour around  $s = 1$ . In Section 5, we carry out the local Hankel analysis via the substitution  $w = (s-1)\log x$  and evaluate the main and lower-order contributions. In Section 6, we extract the asymptotic formula for  $T_j(x)$  from that of  $A(x, y)$  by taking the  $j$ -th derivative at  $y = 0$ , thereby completing the proof of Theorem 1.3.

Finally, in Section 7, we explain why the fixed- $j$  asymptotics do not determine the full alternating sum in the decomposition of  $D(x)$ , and indicate the need for estimates uniform in  $j$  or for a direct analysis near  $y = -1$ .

## 2 Proof of the identity

In this section, we prove the identity stated in Theorem 1.1. The proof proceeds by partitioning the sum  $D(x) = \sum_{n \leq x} d(n)$  according to the value of  $\omega(n)$ , and then applying an inclusion–exclusion argument to the  $\omega(n) \geq 2$  part.

*Proof of Theorem 1.1.* For fixed  $x$ , all sums below are finite. Indeed,  $\pi(x^{1/i}) = 0$  for all sufficiently large  $i$ , and  $\binom{\omega(n)}{j} = 0$  whenever  $j > \omega(n)$ .

Partition the integers  $n$  with  $1 \leq n \leq x$  into three disjoint classes:

- (i)  $n = 1$  (equivalently,  $\omega(n) = 0$ );
- (ii) prime powers, i.e.,  $\omega(n) = 1$ ;
- (iii) integers with  $\omega(n) \geq 2$ .

Writing  $D(x)$  as the sum of contributions from each class, we have

$$D(x) = d(1) + \sum_{\substack{n \leq x \\ \omega(n)=1}} d(n) + \sum_{\substack{n \leq x \\ \omega(n) \geq 2}} d(n). \quad (2.1)$$

Since  $d(1) = 1$ , the first term equals 1.

For the second term, every integer  $n$  with  $\omega(n) = 1$  is of the form  $n = p^i$  for a unique prime  $p$  and a unique integer  $i \geq 1$ , and  $d(p^i) = i + 1$ . The number of prime powers  $p^i \leq x$  with exponent exactly  $i$  equals  $\pi(x^{1/i})$ . Hence

$$\sum_{\substack{n \leq x \\ \omega(n)=1}} d(n) = \sum_{i \geq 1} (i+1) \pi(x^{1/i}). \quad (2.2)$$

For the third term, we apply inclusion–exclusion on the set of distinct prime divisors of  $n$ . For each integer  $j \geq 2$ , recall the definition

$$T_j(x) = \sum_{n \leq x} d(n) \binom{\omega(n)}{j}.$$

We claim that

$$\sum_{\substack{n \leq x \\ \omega(n) \geq 2}} d(n) = \sum_{j \geq 2} (-1)^j (j-1) T_j(x). \quad (2.3)$$

To prove (2.3), interchange the order of summation on the right-hand side. This involves only finite sums, so no convergence issue occurs:

$$\sum_{j \geq 2} (-1)^j (j-1) T_j(x) = \sum_{n \leq x} d(n) \sum_{j \geq 2} (-1)^j (j-1) \binom{\omega(n)}{j}.$$

For  $n$  with  $\omega(n) \in \{0, 1\}$ , we have  $\binom{\omega(n)}{j} = 0$  for all  $j \geq 2$ , so such  $n$  do not contribute. For  $n$  with  $\omega(n) = m \geq 2$ , we use the following combinatorial identity.

**Lemma 2.1.** For every integer  $m \geq 2$ ,

$$\sum_{j=2}^m (-1)^j (j-1) \binom{m}{j} = 1. \quad (2.4)$$

*Proof of the lemma.* Starting from the standard identities

$$\sum_{j=0}^m (-1)^j \binom{m}{j} = 0 \quad \text{and} \quad \sum_{j=0}^m (-1)^j j \binom{m}{j} = \frac{d}{dx} (1-x)^m \Big|_{x=1} = 0 \quad (\text{for } m \geq 2),$$

we combine them to obtain

$$\sum_{j=0}^m (-1)^j (j-1) \binom{m}{j} = 0 \quad (\text{for } m \geq 2).$$

Separating the  $j = 0$  and  $j = 1$  terms:

$$1 \cdot (-1) + 0 \cdot \binom{m}{1} + \sum_{j=2}^m (-1)^j (j-1) \binom{m}{j} = 0,$$

which yields

$$\sum_{j=2}^m (-1)^j (j-1) \binom{m}{j} = 1. \quad \square$$

Applying the lemma, we obtain

$$\sum_{n \leq x} d(n) \sum_{j \geq 2} (-1)^j (j-1) \binom{\omega(n)}{j} = \sum_{\substack{n \leq x \\ \omega(n) \geq 2}} d(n) \cdot 1,$$

which establishes (2.3).

Combining the contributions from all three classes yields

$$D(x) = 1 + \sum_{i \geq 1} (i+1) \pi(x^{1/i}) + \sum_{j \geq 2} (-1)^j (j-1) T_j(x),$$

completing the proof.  $\square$

### 3 Dirichlet series and factorization

In this section, we introduce the two-variable Dirichlet series  $F(s, y)$  and establish its key analytic properties. The main result is the factorization  $F(s, y) = \zeta(s)^{2+2y} H(s, y)$ , where  $H(s, y)$  is holomorphic and uniformly bounded for  $s$  near 1 and  $y$  near 0.

#### 3.1 Euler product

Recall the definition

$$F(s, y) := \sum_{n \geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s}. \quad (3.1)$$

Since the arithmetic function  $n \mapsto d(n)(1+y)^{\omega(n)}$  is multiplicative in  $n$  for each fixed  $y$ , the series admits an Euler product representation for  $\operatorname{Re}(s) > 1$  [1].

**Lemma 3.1.** For  $\operatorname{Re}(s) > 1$  and every  $y \in \mathbb{C}$ ,

$$F(s, y) = \prod_p (1 + (1+y)((1-p^{-s})^{-2} - 1)), \quad (3.2)$$

where the product is over all primes  $p$ .

*Proof.* By multiplicativity, for  $\operatorname{Re}(s) > 1$  we have

$$F(s, y) = \sum_{n \geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s} = \prod_p \left( \sum_{k \geq 0} \frac{d(p^k)(1+y)^{\omega(p^k)}}{p^{ks}} \right).$$

For  $k = 0$ , the summand equals 1. For  $k \geq 1$ , we have  $d(p^k) = k + 1$  and  $\omega(p^k) = 1$ , so the local factor becomes

$$1 + (1+y) \sum_{k \geq 1} \frac{k+1}{p^{ks}}.$$

Setting  $u := p^{-s}$  and using the standard identity

$$\sum_{k \geq 1} (k+1)u^k = \frac{1}{(1-u)^2} - 1 \quad \text{for } |u| < 1,$$

the local factor equals  $1 + (1+y)((1-p^{-s})^{-2} - 1)$ , which proves (3.2).  $\square$

#### 3.2 Factorization isolating the singularity

The Riemann zeta function admits the Euler product [1]  $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$  for  $\operatorname{Re}(s) > 1$ , so that  $\zeta(s)^{2+2y} = \prod_p (1 - p^{-s})^{-(2+2y)}$  on the same half-plane, provided a branch of the logarithm is fixed. We take these powers using the principal branch of the logarithm. With this convention, each local factor is analytic in  $s$  for  $\operatorname{Re}(s) > 1$  and in  $y$  near 0.

Define

$$H(s, y) := \prod_p (1 - p^{-s})^{2+2y} (1 + (1+y)((1-p^{-s})^{-2} - 1)). \quad (3.3)$$

**Corollary 3.2.** For  $\operatorname{Re}(s) > 1$  and  $y$  in a neighborhood of 0,

$$F(s, y) = \zeta(s)^{2+2y} H(s, y). \quad (3.4)$$

*Proof.* This follows by combining Lemma 3.1 with the Euler product for  $\zeta(s)^{2+2y}$  and regrouping factors.  $\square$

The purpose of the factorization (3.4) is to isolate the singular behavior of  $F(s, y)$  at  $s = 1$  in the factor  $\zeta(s)^{2+2y}$ , leaving  $H(s, y)$  as a well-behaved function in a larger region, as we now establish.

### 3.3 Analytic properties of $H(s, y)$

We show that the Euler product (3.3) defines a holomorphic function on a strip containing  $\operatorname{Re}(s) = 1$ , uniformly for  $y$  in a small disk around 0. The key is that each local factor  $H_p(s, y)$  deviates from 1 by a quantity of order  $p^{-2\operatorname{Re}(s)}$ .

**Lemma 3.3.** Fix  $\eta \in (0, 1/2)$  and  $\delta \in (0, 1/2)$ . Write  $u := p^{-s}$  and denote the local factor by

$$H_p(s, y) := (1 - u)^{2+2y} (1 + (1 + y) ((1 - u)^{-2} - 1)).$$

Then for  $|y| \leq \eta$  and  $\operatorname{Re}(s) \geq 1 - \delta$ ,

$$H_p(s, y) = 1 + O_{\eta, \delta}(u^2), \quad (3.5)$$

where the implied constant depends only on  $\eta$  and  $\delta$ .

*Proof.* The condition  $\operatorname{Re}(s) \geq 1 - \delta$  implies that, for every prime  $p \geq 2$ ,

$$|u| = p^{-\operatorname{Re}(s)} \leq p^{-(1-\delta)} \leq 2^{-(1-\delta)} < 1,$$

since  $\delta < 1/2$ . In particular,  $|u|$  is bounded away from 1, and the power series expansions

$$(1 - u)^{-2} - 1 = 2u + 3u^2 + 4u^3 + \cdots = 2u + O_\delta(u^2),$$

$$(1 - u)^{2+2y} = 1 - (2 + 2y)u + \binom{2 + 2y}{2} u^2 + \cdots = 1 - (2 + 2y)u + O_{\eta, \delta}(u^2),$$

converge absolutely, with remainders uniform for  $|y| \leq \eta$  and  $\operatorname{Re}(s) \geq 1 - \delta$ . Multiplying the expansions:

$$H_p(s, y) = (1 - (2 + 2y)u + O_{\eta, \delta}(u^2)) (1 + (1 + y)(2u + O_\delta(u^2))).$$

The coefficient of  $u^1$  on the right is  $-(2 + 2y) + 2(1 + y) = 0$ . Hence  $H_p(s, y) = 1 + O_{\eta, \delta}(u^2)$ , proving (3.5).  $\square$

**Proposition 3.4.** Fix  $\eta \in (0, 1/2)$  and  $\delta \in (0, 1/2)$ , and let

$$\mathcal{R}_{\delta, \eta} := \{(s, y) \in \mathbb{C}^2 : \operatorname{Re}(s) \geq 1 - \delta, |y| \leq \eta\}. \quad (3.6)$$

Then the partial products of the Euler product (3.3) converge absolutely and uniformly on  $\mathcal{R}_{\delta, \eta}$  to a function  $H(s, y)$ . For every  $y$  with  $|y| \leq \eta$ , the function  $H(s, y)$  is holomorphic in  $s$  on the open strip  $\operatorname{Re}(s) > 1 - \delta$ . Moreover,  $H(s, y)$  is jointly holomorphic in  $(s, y)$  on  $\operatorname{int}(\mathcal{R}_{\delta, \eta})$ , and  $H$  is uniformly bounded on  $\mathcal{R}_{\delta, \eta}$ .

*Proof.* By Lemma 3.3, for every  $(s, y) \in \mathcal{R}_{\delta, \eta}$  and every prime  $p$ ,

$$|H_p(s, y) - 1| \leq C_{\eta, \delta} p^{-2(1-\delta)}, \quad (3.7)$$

where the constant  $C_{\eta,\delta}$  depends only on  $\eta$  and  $\delta$ . Since  $\delta < 1/2$ , we have  $2(1-\delta) > 1$ , and the series  $\sum_n n^{-2(1-\delta)}$  converges; in particular,  $\sum_p p^{-2(1-\delta)}$  converges. By the Weierstrass  $M$ -test, the series

$$\sum_p |H_p(s, y) - 1|$$

converges uniformly on  $\mathcal{R}_{\delta,\eta}$ . Hence the infinite product criterion applies: if  $\sum_n |u_n|$  converges uniformly, then the product  $\prod_n (1 + u_n)$  converges absolutely and uniformly. Therefore the infinite product  $\prod_p H_p(s, y)$  converge absolutely and uniformly on  $\mathcal{R}_{\delta,\eta}$ . In particular, the partial products  $\prod_{p \leq P} H_p(s, y)$  converge uniformly to a limit, which we denote by  $H(s, y)$ .

To deduce holomorphy in  $s$ , fix  $y_0 \in \mathbb{C}$  with  $|y_0| \leq \eta$  and let  $s_0 \in \mathbb{C}$  be any point with  $\operatorname{Re}(s_0) > 1 - \delta$ . Choose  $r > 0$  small enough that  $r < \operatorname{Re}(s_0) - (1 - \delta)$ , and define

$$K := \overline{D}(s_0, r) = \{s \in \mathbb{C} : |s - s_0| \leq r\}. \quad (3.8)$$

The set  $K$  is closed and bounded in  $\mathbb{C}$ , so by the Heine–Borel theorem it is compact. By the choice of  $r$ , every  $s \in K$  satisfies

$$\operatorname{Re}(s) \geq \operatorname{Re}(s_0) - r > 1 - \delta,$$

and hence  $K \times \{y_0\} \subset \mathcal{R}_{\delta,\eta}$ . The uniform convergence of the partial products on  $\mathcal{R}_{\delta,\eta}$  established above therefore restricts to uniform convergence of  $\prod_{p \leq P} H_p(s, y_0)$  on  $K$  as  $P \rightarrow \infty$ .

Each finite partial product  $\prod_{p \leq P} H_p(s, y_0)$  is holomorphic in  $s$  on the open disk  $D(s_0, r) = \operatorname{int}(K)$ , since each factor  $H_p(\cdot, y_0)$  is. By the theorem on uniform limits of holomorphic functions (Weierstrass), the limit  $H(\cdot, y_0)$  is holomorphic on  $D(s_0, r)$ , and in particular at  $s_0$ . Since  $s_0$  with  $\operatorname{Re}(s_0) > 1 - \delta$  was arbitrary,  $H(\cdot, y_0)$  is holomorphic on the open strip  $\operatorname{Re}(s) > 1 - \delta$ . Since  $y_0$  with  $|y_0| \leq \eta$  was also arbitrary, the holomorphy claim in  $s$  is proved.

For joint holomorphy, each finite partial product  $\prod_{p \leq P} H_p(s, y)$  is holomorphic in  $(s, y)$  on  $\operatorname{int}(\mathcal{R}_{\delta,\eta})$ , since each factor  $H_p(s, y)$  is. The uniform convergence of the partial products on  $\mathcal{R}_{\delta,\eta}$  established above restricts to uniform convergence on every compact subset of  $\operatorname{int}(\mathcal{R}_{\delta,\eta})$ , since  $\operatorname{int}(\mathcal{R}_{\delta,\eta}) \subset \mathcal{R}_{\delta,\eta}$ . By the theorem on uniform limits of holomorphic functions in two complex variables, the limit  $H(s, y)$  is jointly holomorphic on  $\operatorname{int}(\mathcal{R}_{\delta,\eta})$ .

Uniform boundedness on  $\mathcal{R}_{\delta,\eta}$  follows from (3.7):

$$|H(s, y)| \leq \prod_p (1 + |H_p(s, y) - 1|) \leq \exp \left( \sum_p |H_p(s, y) - 1| \right) \leq \exp \left( C_{\eta,\delta} \sum_p p^{-2(1-\delta)} \right),$$

the right-hand side being finite and independent of  $(s, y) \in \mathcal{R}_{\delta,\eta}$ . □

**Lemma 3.5.** The function  $H(s, y)$  satisfies

$$H(s, 0) \equiv 1 \quad \text{on } \operatorname{Re}(s) > 1/2. \quad (3.9)$$

In particular,  $H(1, 0) = 1$ .

*Proof.* On  $\operatorname{Re}(s) > 1$ , the factorization (3.4) specialized at  $y = 0$  reads

$$F(s, 0) = \zeta(s)^2 H(s, 0).$$

On the other hand, the Dirichlet series defining  $F$  gives

$$F(s, 0) = \sum_{n \geq 1} \frac{d(n)}{n^s} = \zeta(s)^2 \quad \text{on } \operatorname{Re}(s) > 1.$$



Comparing the two expressions yields  $\zeta(s)^2 H(s, 0) = \zeta(s)^2$  on  $\operatorname{Re}(s) > 1$ . Since  $\zeta(s) \neq 0$  on  $\operatorname{Re}(s) > 1$  (which follows directly from the Euler product representation of  $\zeta$ ), we may divide both sides by  $\zeta(s)^2$  to obtain

$$H(s, 0) \equiv 1 \quad \text{on } \operatorname{Re}(s) > 1.$$

By Proposition 3.4, applied with any  $\delta \in (0, 1/2)$ , the function  $H(s, 0)$  is holomorphic on  $\operatorname{Re}(s) > 1 - \delta$ ; letting  $\delta \rightarrow 1/2$  shows that  $H(s, 0)$  is holomorphic on the connected open set  $\operatorname{Re}(s) > 1/2$ . The constant function 1 is also holomorphic there, and the two functions agree on the open subset  $\operatorname{Re}(s) > 1$ . By the identity theorem, they agree on all of  $\operatorname{Re}(s) > 1/2$ . In particular,  $H(1, 0) = 1$ .  $\square$

## 4 Perron formula and contour deformation

To extract the asymptotic behavior of  $T_j(x)$  from the Dirichlet series  $F(s, y)$ , we work with the generating partial sum

$$A(x, y) := \sum_{n \leq x} d(n)(1+y)^{\omega(n)}, \quad (4.1)$$

which satisfies the generating relation

$$A(x, y) = \sum_{j \geq 0} T_j(x) y^j. \quad (4.2)$$

We abbreviate  $\alpha = 2 + 2y$ .

We treat the parameters  $\eta, \delta \in (0, 1/2)$  that appeared as local parameters in Lemma 3.3 and Proposition 3.4 as global constants for the remainder of the paper. The entire analysis of Section 4 takes place within the region  $\mathcal{R}_{\delta, \eta}$ , on which the conclusions of Proposition 3.4 apply to  $H(s, y)$ .

In Section 4.3 we make a specific choice  $\delta' \in (0, 1/2)$  of  $\delta$ , constrained by zero-free region constant  $c_0$ , together with a truncation-dependent  $\lambda_T < \delta'$ . Since  $\delta'$  is a particular admissible value of  $\delta$ , the conclusions of Proposition 3.4 hold on  $\mathcal{R}_{\delta', \eta}$ . After  $\delta'$  is fixed, we may shrink  $\eta$  if necessary for the contour estimates.

For  $|y| \leq \eta$ , the exponent  $\alpha = 2 + 2y$  lies in the disk of radius  $2\eta$  centered at 2; since  $2\eta < 1$ , it is bounded away from non-positive integers.

### 4.1 The truncated Perron formula

**Theorem 4.1** (Truncated Perron formula [6]). Let  $f(s) = \sum_{n \geq 1} a_n n^{-s}$  be a Dirichlet series with abscissa of absolute convergence  $\sigma_a$ . For real numbers  $c > \sigma_a$ ,  $x \geq 2$ , and  $T \geq 1$  with  $x \notin \mathbb{Z}$ ,

$$\sum_{n \leq x} a_n = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} f(s) \frac{x^s}{s} ds + R(x, T), \quad (4.3)$$

where the truncation error satisfies

$$R(x, T) \ll x^c \sum_{n \geq 1} \frac{|a_n|}{n^c} \frac{1}{1 + T \left| \log \frac{x}{n} \right|}. \quad (4.4)$$

If  $x = N$  is an integer, we do not apply the Perron formula directly at  $N$ . Instead, we apply it at

$$X := N + \frac{1}{2}.$$

Then  $X \notin \mathbb{Z}$  and

$$A(N, y) = A(X, y),$$

because the integers satisfying  $n \leq N$  are exactly the integers satisfying  $n \leq N + 1/2$ . Moreover,

$$X = N + O(1), \quad \log X = \log N + O(1/N),$$

so every asymptotic estimate obtained for  $X$  transfers back to  $N$ . Thus it is enough in the Perron argument to work with non-integral  $x$ ; the integer case follows from this replacement.

We apply Theorem 4.1 to the Dirichlet series  $F(s, y) = \sum_{n \geq 1} d(n)(1+y)^{\omega(n)} n^{-s}$ . For  $|y| \leq \eta$  with  $\eta$  sufficiently small, the series converges absolutely for  $\operatorname{Re}(s) > 1$ , so we may take any  $c > 1$ .

For  $x \geq 2$  with  $x \notin \mathbb{Z}$ ,  $|y| \leq \eta$ , and  $T \geq 1$ , put  $c = 1 + 1/\log x$ . Theorem 4.1 gives

$$A(x, y) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s, y) \frac{x^s}{s} ds + E_{\text{Perron}}(x, T, y), \quad (4.5)$$

where

$$E_{\text{Perron}}(x, T, y) \ll_{\eta} x^c \sum_{n \geq 1} \frac{d(n)(1+\eta)^{\omega(n)}}{n^c} \frac{1}{1 + T \left| \log \frac{x}{n} \right|}. \quad (4.6)$$

Indeed, applying (4.4) with  $a_n = d(n)(1+y)^{\omega(n)}$  and using  $|1+y| \leq 1+\eta$  gives the displayed bound.

## 4.2 Analytic preliminaries

To control the contour integrals arising from the deformation in Section 4.3, we will need:

- (a) a region in which  $\zeta(s)$  does not vanish, so that the contour can avoid its zeros;
- (b) a continuous branch of  $\zeta(s)^{2+2y}$  on a neighborhood of the contour;
- (c) a uniform bound on  $|\zeta(s)|$  along vertical lines in the critical strip;
- (d) a corresponding bound on  $\arg \zeta(s)$  on the chosen branch.

We collect these analytic ingredients here, before constructing the contour itself in Section 4.3.

### Zero-free region

For the contour deformation to be valid, we need a region in which  $\zeta(s) \neq 0$ . We use the following classical result of de la Vallée Poussin, quoted without proof.

**Theorem 4.2** (de la Vallée Poussin zero-free region [7]). There exists an absolute constant  $c_0 \in (0, 1/2)$  such that  $\zeta(s) \neq 0$  on the open region

$$\mathcal{Z} := \left\{ s = \sigma + it \in \mathbb{C} : \sigma > 1 - \frac{c_0}{\log(|t| + 2)} \right\} \setminus \{1\}. \quad (4.7)$$

The boundary of  $\mathcal{Z}$  is curved, and integrating along it would complicate the estimation of the resulting contour pieces. We will instead use a fixed vertical line lying strictly inside  $\mathcal{Z}$  for all  $|t| \leq T$ ; the precise depth of this line is fixed in Section 4.3.

### Continuous branch of $\zeta(s)^\alpha$

To make sense of  $\zeta(s)^\alpha = \zeta(s)^{2+2y}$  on a neighborhood of the contour, we specify a branch on the region

$$\mathcal{D} := \mathcal{Z} \setminus \mathcal{S}, \quad (4.8)$$

where  $\mathcal{S} := \{s \in \mathbb{C} : \Im(s) = 0, \operatorname{Re}(s) \leq 1\}$  is the slit along the real axis to the left of  $s = 1$ . Cutting  $\mathcal{Z}$  along  $\mathcal{S}$  breaks the only loop around the puncture  $s = 1$ , so that  $\mathcal{D}$  is simply connected.<sup>1</sup>

On  $\mathcal{D}$ , both  $\zeta(s)$  and  $\zeta'(s)$  are holomorphic, and  $\zeta(s) \neq 0$ . Fix any real number  $c_* > 1$  (for example  $c_* = 2$ ); since  $\zeta(c_*) > 0$ , the principal logarithm  $\log \zeta(c_*) \in \mathbb{R}$  is well-defined. Define

$$L(s) := \log \zeta(c_*) + \int_{c_*}^s \frac{\zeta'(\omega)}{\zeta(\omega)} d\omega \quad (s \in \mathcal{D}), \quad (4.9)$$

where the integral is taken along any path in  $\mathcal{D}$  from  $c_*$  to  $s$ . Since  $\mathcal{D}$  is simply connected and  $\zeta'/\zeta$  is holomorphic on  $\mathcal{D}$ , the integral is path-independent, and  $L$  is a well-defined holomorphic function on  $\mathcal{D}$ .

We claim  $e^{L(s)} = \zeta(s)$  on  $\mathcal{D}$ . Indeed,  $L'(s) = \zeta'(s)/\zeta(s)$  on  $\mathcal{D}$ , so

$$\frac{d}{ds}(e^{-L(s)}\zeta(s)) = e^{-L(s)}(\zeta'(s) - L'(s)\zeta(s)) = 0.$$

Hence  $e^{-L(s)}\zeta(s)$  is constant on  $\mathcal{D}$ , and evaluating at  $s = c_*$  gives  $e^{-L(c_*)}\zeta(c_*) = e^{-\log \zeta(c_*)}\zeta(c_*) = 1$ . Therefore  $e^{L(s)} = \zeta(s)$  throughout  $\mathcal{D}$ , so  $L$  serves as a continuous branch of  $\log \zeta$  on  $\mathcal{D}$ .

We define

$$\zeta(s)^\alpha := e^{\alpha L(s)} \quad (s \in \mathcal{D}, \alpha \in \mathbb{C}). \quad (4.10)$$

This is single-valued and holomorphic in  $s$  on  $\mathcal{D}$ , and entire in  $\alpha$ . At  $\alpha = 2$ , the definition reduces to  $\zeta(s)^2$  in the usual sense.

### Vertical growth of $\zeta(s)$

We use the following uniform bound on  $|\zeta(s)|$ , obtained by combining the convexity bound with the trivial estimate  $\zeta(\sigma + it) \ll \log |t|$  for  $\sigma \geq 1$ .

**Theorem 4.3** (Uniform bound on  $\zeta(s)$  in a vertical strip [7]). Fix  $\delta \in (0, 1/2)$ . For every  $\varepsilon > 0$ ,

$$|\zeta(\sigma + it)| \ll_{\delta, \varepsilon} |t|^{\delta/2 + \varepsilon} \quad \text{for } 1 - \delta \leq \sigma \leq 1 + \delta \text{ and } |t| \geq 2, \quad (4.11)$$

where the implied constant is independent of  $t$ .

*Sketch.* For  $1 - \delta \leq \sigma \leq 1$ , the convexity bound gives  $|\zeta(\sigma + it)| \ll_\varepsilon |t|^{(1-\sigma)/2 + \varepsilon}$ . Taking the supremum of  $(1 - \sigma)/2$  over  $\sigma \in [1 - \delta, 1]$  yields  $\delta/2$ , so  $|\zeta(\sigma + it)| \ll_\varepsilon |t|^{\delta/2 + \varepsilon}$  on this subrange.

For  $1 \leq \sigma \leq 1 + \delta$ , the standard estimate  $|\zeta(\sigma + it)| \ll \log |t|$  holds, and  $\log |t| \ll_\varepsilon |t|^\varepsilon \leq |t|^{\delta/2 + \varepsilon}$ . Combining the two ranges gives (4.11).  $\square$

<sup>1</sup>Removing the slit from  $s = 1$  to the boundary cuts the only possible loop around the puncture  $s = 1$ , just as  $\mathbb{C} \setminus (-\infty, 0]$  is the standard simply connected branch domain for the logarithm. This is the topological input needed for path independence in the definition of  $L(s)$ .

### Bound on $\arg \zeta$

We also use the following standard estimate on the branch determined by (4.9).

**Theorem 4.4** (Bound on  $\arg \zeta$  [7]). There exists an absolute constant  $C_{\arg} > 0$  such that, for every  $s = \sigma + it \in \mathcal{D}$  with  $|t| \geq 2$ ,

$$|\arg \zeta(s)| \leq C_{\arg} \log |t|, \quad (4.12)$$

where the branch of  $\arg \zeta$  is the one determined by the logarithm  $L$  on  $\mathcal{D}$ .

### 4.3 Contour decomposition

We now introduce the parameters governing the contour deformation and construct the deformed contour.

#### Depth of the deformation

Fix once and for all an absolute constant

$$\delta' \in \left( \frac{c_0}{2 \log 3}, \frac{c_0}{2} \right), \quad (4.13)$$

where  $c_0$  is the constant of Theorem 4.2. This range of  $\delta'$  is chosen so that the deformed contour remains in the fixed strip used in Section 4.4.

For each truncation height  $T \geq 1$ , define

$$\lambda_T := \frac{c_0}{2 \log(T+2)}. \quad (4.14)$$

By construction,  $\lambda_T \rightarrow 0$  as  $T \rightarrow \infty$ , and for every  $T \geq 1$ ,

$$\lambda_T \leq \frac{c_0}{2 \log 3} < \delta', \quad (4.15)$$

so the depth  $\sigma_0(T) := 1 - \lambda_T$  of the deformed contour satisfies  $\sigma_0(T) > 1 - \delta'$ . Consequently, the contour  $\mathcal{C}(T, \rho)$  defined below lies in  $\mathcal{R}_{\delta', \eta}$  for every  $T \geq 1$  and every  $|y| \leq \eta$ . In particular, Proposition 3.4 applies on the contour with  $\delta = \delta'$ .

Finally, the choice  $c = 1 + 1/\log x$  in the Perron representation (4.5) satisfies  $c \leq 1 + \delta'$  for all  $x$  sufficiently large, ensuring that the abscissa  $c$  lies inside the strip  $1 - \delta' \leq \sigma \leq 1 + \delta'$  used in the contour estimates. We assume  $x$  is large enough for this throughout.

#### The deformed contour

Fix a truncation height  $T \geq 1$ , and set

$$\sigma_0 := \sigma_0(T) := 1 - \lambda_T. \quad (4.16)$$

Choose a parameter  $\rho > 0$  with  $\rho < \lambda_T$ ; its precise choice as a function of  $x$  will be made in Section 5. The notation  $i0^\pm$  below denotes boundary values: the two Hankel arms are first taken at heights  $\pm \varepsilon$  with  $\varepsilon > 0$ , and the limit  $\varepsilon \rightarrow 0^+$  is taken after the corresponding integrals are formed. Thus, for each  $\varepsilon > 0$ , the approximating contour lies in the slit domain  $\mathcal{D}$ , and the limit  $\varepsilon \rightarrow 0^+$  is taken only afterwards.

We deform the original Perron segment  $[c - iT, c + iT]$  leftward, around the singularity at  $s = 1$ , into a contour  $\mathcal{C} = \mathcal{C}(T, \rho)$  traversed from  $c - iT$  to  $c + iT$ . It consists of the following five pieces:

- (1)  $\mathcal{C}_{\text{bot}}$ : the horizontal segment from  $c - iT$  to  $\sigma_0 - iT$ .
- (2)  $\mathcal{C}_{\text{left}}^-$ : the vertical segment from  $\sigma_0 - iT$  to  $\sigma_0 + i0^-$ .
- (3)  $\mathcal{H}_{\text{loc}}$ : a local Hankel contour around  $s = 1$ , consisting of
  - (3a) the lower arm from  $\sigma_0 + i0^-$  to  $1 - \rho + i0^-$ ;
  - (3b) the circle  $|s - 1| = \rho$ , traversed from  $1 - \rho + i0^-$  to  $1 - \rho + i0^+$  counterclockwise through the right side of  $s = 1$ ;
  - (3c) the upper arm from  $1 - \rho + i0^+$  to  $\sigma_0 + i0^+$ .
- (4)  $\mathcal{C}_{\text{left}}^+$ : the vertical segment from  $\sigma_0 + i0^+$  to  $\sigma_0 + iT$ .
- (5)  $\mathcal{C}_{\text{top}}$ : the horizontal segment from  $\sigma_0 + iT$  to  $c + iT$ .

The five pieces are joined consecutively, so  $\mathcal{C}$  has the same initial and terminal points as the original Perron contour. The Hankel arms are understood as boundary values along the slit  $\mathcal{S}$ ; in the local representation near  $s = 1$ , their discontinuity is carried by the factor  $(s - 1)^{-2-2y}$  and produces the local contribution  $I_H$ .

### Verification that the contour avoids $\zeta$ -zeros

We claim that  $\mathcal{C} \setminus \{1\} \subset \mathcal{Z}$ , where  $\mathcal{Z}$  is the zero-free region of Theorem 4.2. For the horizontal pieces  $\mathcal{C}_{\text{bot}}$  and  $\mathcal{C}_{\text{top}}$ ,

$$\operatorname{Re}(s) \geq \sigma_0(T) = 1 - \frac{c_0}{2 \log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \quad (4.17)$$

For the vertical pieces  $\mathcal{C}_{\text{left}}^\pm$ ,

$$\operatorname{Re}(s) = \sigma_0(T) = 1 - \frac{c_0}{2 \log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \quad (4.18)$$

For the local Hankel contour  $\mathcal{H}_{\text{loc}}$ , the two horizontal arms (3a) and (3c) satisfy  $\operatorname{Re}(s) \geq \sigma_0(T)$  and  $|\Im s| \leq \rho$ , hence are covered by the same estimate as the vertical pieces. For the small circle (3b),

$$\operatorname{Re}(s) \geq 1 - \rho > 1 - \lambda_T = 1 - \frac{c_0}{2 \log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \quad (4.19)$$

Therefore  $\mathcal{C} \setminus \{1\} \subset \mathcal{Z}$ . Combined with the boundary-value convention for the Hankel arms, the approximating contours with heights  $\pm \varepsilon$  are contained in  $\mathcal{D}$  away from  $s = 1$ . The integrand  $F(s, y)x^s/s$  is therefore well-defined and holomorphic on a neighborhood of each approximating contour, and the contour  $\mathcal{C}$  is understood as the corresponding boundary-value limit.

### Cauchy's theorem

For  $\varepsilon > 0$ , replace the two Hankel arms along the slit by parallel paths at heights  $-\varepsilon$  and  $+\varepsilon$ , and denote the resulting deformed contour by  $\mathcal{C}_\varepsilon$ . The Perron contour and  $\mathcal{C}_\varepsilon$  share the same endpoints  $c \pm iT$ . Together, the Perron contour traversed from  $c - iT$  to  $c + iT$  and  $\mathcal{C}_\varepsilon$  traversed in the reverse direction form a closed loop  $\Gamma_\varepsilon$ , oriented counterclockwise. The keyhole formed by the approximating Hankel contour and the slit  $\mathcal{S}$  excludes the point  $s = 1$  from the interior of  $\Gamma_\varepsilon$ , so the interior of  $\Gamma_\varepsilon$  is contained in the rectangle

$$\mathcal{R}_\Gamma := \{s = \sigma + it : \sigma_0(T) \leq \sigma \leq c, |t| \leq T\} \setminus \mathcal{S}. \quad (4.20)$$

For every  $s \in \mathcal{R}_\Gamma$ ,

$$\operatorname{Re}(s) \geq \sigma_0(T) = 1 - \frac{c_0}{2\log(T+2)} > 1 - \frac{c_0}{\log(|\Im s|+2)}, \quad (4.21)$$

so  $\mathcal{R}_\Gamma \subset \mathcal{Z}$ . Combined with the on-contour verifications (4.17)–(4.19), this shows that  $\Gamma_\varepsilon$  together with its interior, with the slit removed, is contained in the slit domain  $\mathcal{D}$ . Hence  $F(s, y)x^s/s$  is holomorphic on an open neighborhood of  $\Gamma_\varepsilon$  and its interior. By Cauchy's theorem,

$$\oint_{\Gamma_\varepsilon} F(s, y) \frac{x^s}{s} ds = 0, \quad (4.22)$$

which yields

$$\int_{c-iT}^{c+iT} F(s, y) \frac{x^s}{s} ds = \int_{\mathcal{C}_\varepsilon} F(s, y) \frac{x^s}{s} ds.$$

Letting  $\varepsilon \rightarrow 0^+$  gives the boundary-value contour  $\mathcal{C}$  and hence

$$\int_{c-iT}^{c+iT} F(s, y) \frac{x^s}{s} ds = \int_{\mathcal{C}} F(s, y) \frac{x^s}{s} ds. \quad (4.23)$$

### Decomposition of $A(x, y)$

For non-integral  $x$ , combining (4.23) with the Perron representation (4.5), we obtain

$$A(x, y) = I_H + I_{\text{top}} + I_{\text{bot}} + I_{\text{left}} + E_{\text{Perron}}(x, T, y), \quad (4.24)$$

where, for  $* \in \{\text{top}, \text{bot}, \text{left}\}$ ,

$$I_* := \frac{1}{2\pi i} \int_{\mathcal{C}_*} F(s, y) \frac{x^s}{s} ds, \quad (4.25)$$

with  $\mathcal{C}_{\text{left}} := \mathcal{C}_{\text{left}}^- \cup \mathcal{C}_{\text{left}}^+$ , and

$$I_H := \frac{1}{2\pi i} \int_{\mathcal{H}_{\text{loc}}} F(s, y) \frac{x^s}{s} ds. \quad (4.26)$$

We write  $\mathcal{H}_{\text{loc}} = \mathcal{H}_{\text{loc}}^- \cup \mathcal{H}_{\text{loc}}^\circ \cup \mathcal{H}_{\text{loc}}^+$  for the lower arm, the small circle, and the upper arm, respectively. The lower arm is oriented from  $\sigma_0 + i0^-$  to  $1 - \rho + i0^-$ , the small circle is oriented counterclockwise from  $1 - \rho + i0^-$  to  $1 - \rho + i0^+$  through the right side of  $s = 1$ , and the upper arm is oriented from  $1 - \rho + i0^+$  to  $\sigma_0 + i0^+$ .

The local Hankel contribution  $I_H$  contains the main asymptotic behavior and will be evaluated in Section 5. The remaining contributions  $I_{\text{top}}, I_{\text{bot}}, I_{\text{left}}$  are estimated in Section 4.4 and shown to be of lower order than the main term, after a suitable choice of  $T = T(x)$ .

## 4.4 Estimates of the contour pieces

In this section, we estimate the three nonlocal contour contributions  $I_{\text{top}}, I_{\text{bot}}, I_{\text{left}}$  appearing in the decomposition (4.24), and combine them with the Perron truncation error  $E_{\text{Perron}}$  to obtain a single error bound. The local Hankel contribution  $I_H$  is deferred to Section 5.

We first collect the two bounds for  $F(s, y)$  used below.

**Lemma 4.5.** Let  $\varepsilon > 0$ . Define

$$B := B(\eta, \delta', \varepsilon) := (2 + 2\eta) \left( \frac{\delta'}{2} + \varepsilon \right) + 2\eta C_{\text{arg}}. \quad (4.27)$$

Then the following estimates hold uniformly for  $|y| \leq \eta$ .

1. If  $s = \sigma + it \in \mathcal{D}$ ,  $|t| \geq 2$ , and  $1 - \delta' \leq \sigma \leq 1 + \delta'$ , then

$$|F(s, y)| \ll_{\eta, \delta', \varepsilon} |t|^B. \quad (4.28)$$

2. If  $T \geq 1$ ,  $s = 1 - \lambda_T + it$ , and  $|t| \leq 2$ , then

$$|F(s, y)| \ll_{\eta, \delta'} \lambda_T^{-2-2\eta} \ll_{\eta, \delta'} (\log(T+2))^{2+2\eta}. \quad (4.29)$$

*Proof.* For the first estimate, let  $s = \sigma + it \in \mathcal{D}$  with  $|t| \geq 2$  and  $1 - \delta' \leq \sigma \leq 1 + \delta'$ . On  $\mathcal{D}$ , the chosen branch gives

$$F(s, y) = \zeta(s)^\alpha H(s, y), \quad \zeta(s)^\alpha = \exp(\alpha L(s)).$$

Writing  $L(s) = \log |\zeta(s)| + i \arg \zeta(s)$  on this branch,

$$|\zeta(s)^\alpha| = |\zeta(s)|^{\operatorname{Re}(\alpha)} \exp(-\Im(\alpha) \arg \zeta(s)).$$

Since  $|y| \leq \eta$ , we have  $\operatorname{Re}(\alpha) \leq 2 + 2\eta$  and  $|\Im(\alpha)| \leq 2\eta$ . Hence

$$|F(s, y)| \leq |H(s, y)| (1 + |\zeta(s)|)^{2+2\eta} \exp(2\eta |\arg \zeta(s)|).$$

Here the factor  $1 + |\zeta(s)|$  avoids any lower-bound assumption on  $|\zeta(s)|$ . Proposition 3.4, applied with  $\delta = \delta'$ , gives  $|H(s, y)| \ll_{\eta, \delta'} 1$ . Moreover, Theorem 4.3 gives

$$|\zeta(s)| \ll_{\delta', \varepsilon} |t|^{\delta'/2+\varepsilon},$$

and Theorem 4.4 gives

$$|\arg \zeta(s)| \leq C_{\arg} \log |t|.$$

Since  $|t| \geq 2$ , these estimates imply

$$|F(s, y)| \ll_{\eta, \delta', \varepsilon} |t|^{(2+2\eta)(\delta'/2+\varepsilon)+2\eta C_{\arg}} = |t|^B,$$

which proves (4.28).

We now prove the low-height estimate. Let

$$q(s) := (s-1)\zeta(s), \quad q(1) := 1.$$

Then  $q$  is holomorphic at  $s = 1$ . Put

$$\lambda_0 := \frac{c_0}{2 \log 3}, \quad K := \{s = \sigma + it : 1 - \lambda_0 \leq \sigma \leq 1, |t| \leq 2\}.$$

For  $|t| \leq 2$  we have  $\log(|t|+2) \leq \log 4 < 2 \log 3$ , and hence

$$\lambda_0 = \frac{c_0}{2 \log 3} < \frac{c_0}{\log(|t|+2)}.$$

Thus  $K \setminus \{1\}$  lies in the zero-free region of Theorem 4.2. After setting  $q(1) = 1$ , the function  $q$  is holomorphic and non-vanishing on  $K$ . Since  $K$  is compact, there is an open neighborhood  $U$  of  $K$  on which  $q$  is non-vanishing. Choose a holomorphic branch

$$M(s) = \log q(s)$$

on  $U$ , compatible with  $M(s) = L(s) + \text{Log}(s-1)$  on  $U \cap \mathcal{D}$ . Let

$$M_K := \sup_{s \in K} |M(s)| < \infty.$$

For  $|y| \leq \eta$ ,

$$|q(s)^\alpha| = |\exp(\alpha M(s))| \leq \exp(|\alpha| |M(s)|) \leq \exp((2+2\eta)M_K) \quad (s \in K).$$

Define

$$G(s, y) := q(s)^\alpha H(s, y). \quad (4.30)$$

Then, on the chosen branch near the slit,

$$F(s, y) = G(s, y)(s-1)^{-\alpha}. \quad (4.31)$$

Since  $K \subset \{s : \text{Re}(s) \geq 1 - \delta'\}$ , combining the preceding bound for  $q(s)^\alpha$  with Proposition 3.4, again with  $\delta = \delta'$ , gives

$$|G(s, y)| = |q(s)^\alpha H(s, y)| \ll_{\eta, \delta'} 1 \quad (s \in K, |y| \leq \eta).$$

On the left vertical line  $s = 1 - \lambda_T + it$  with  $|t| \leq 2$ , we use (4.31). For either boundary value of the logarithm along the slit,  $|\text{Arg}(s-1)| \leq \pi$ , so

$$|(s-1)^{-\alpha}| = |s-1|^{-\text{Re}(\alpha)} \exp(\Im(\alpha) \text{Arg}(s-1)) \ll_\eta |s-1|^{-\text{Re}(\alpha)}.$$

Since  $|s-1| \geq \lambda_T$ , if  $|s-1| \leq 1$  then

$$|s-1|^{-\text{Re}(\alpha)} \leq |s-1|^{-2-2\eta} \leq \lambda_T^{-2-2\eta}.$$

If  $|s-1| \geq 1$ , then  $\text{Re}(\alpha) \geq 2-2\eta > 0$ , so  $|s-1|^{-\text{Re}(\alpha)} \leq 1 \leq \lambda_T^{-2-2\eta}$ . Hence

$$|F(1 - \lambda_T + it, y)| \ll_{\eta, \delta'} \lambda_T^{-2-2\eta}.$$

The definition (4.14) gives

$$\lambda_T^{-2-2\eta} = \left( \frac{2 \log(T+2)}{c_0} \right)^{2+2\eta} \ll_{\eta, \delta'} (\log(T+2))^{2+2\eta},$$

which proves (4.29). □

Since

$$B(\eta, \delta', \varepsilon) = (2+2\eta) \left( \frac{\delta'}{2} + \varepsilon \right) + 2\eta C_{\text{arg}} \longrightarrow \delta' \quad (\eta, \varepsilon \rightarrow 0^+),$$

and  $\delta' < c_0/2$ , we may choose  $\eta > 0$  and  $\varepsilon > 0$  sufficiently small so that

$$B < c_0/2. \quad (4.32)$$

This choice is fixed throughout the contour estimates.



## Nonlocal contour pieces

**Lemma 4.6.** For all sufficiently large  $x$  and every  $T \geq 2$ ,

$$|I_{\text{top}}| + |I_{\text{bot}}| \ll_{\eta, \varepsilon} x^c T^{B-1}, \quad (4.33)$$

uniformly in  $|y| \leq \eta$ .

*Proof.* On  $\mathcal{C}_{\text{top}}$ , parameterize  $s = \sigma + iT$  with  $\sigma_0(T) \leq \sigma \leq c$ . Then

$$\begin{aligned} |I_{\text{top}}| &= \left| \frac{1}{2\pi i} \int_{\mathcal{C}_{\text{top}}} F(s, y) \frac{x^s}{s} ds \right| \\ &\leq \frac{1}{2\pi} \int_{\sigma_0(T)}^c \left| F(\sigma + iT, y) \frac{x^{\sigma+iT}}{\sigma + iT} \right| d\sigma \\ &\leq \frac{c - \sigma_0(T)}{2\pi} \sup_{\sigma_0(T) \leq \sigma \leq c} \left| F(\sigma + iT, y) \frac{x^{\sigma+iT}}{\sigma + iT} \right|. \end{aligned}$$

On this segment,  $|x^{\sigma+iT}| = x^\sigma \leq x^c$  and  $|\sigma + iT| \geq T$ . Also  $\sigma_0(T) \geq 1 - \delta'$  by (4.15), while  $c \leq 1 + \delta'$  for all sufficiently large  $x$ . Hence Lemma 4.5 gives

$$\sup_{\sigma_0(T) \leq \sigma \leq c} |F(\sigma + iT, y)| \ll_{\eta, \varepsilon} T^B.$$

Since

$$c - \sigma_0(T) = \frac{1}{\log x} + \lambda_T = \frac{1}{\log x} + \frac{c_0}{2 \log(T+2)} \ll 1,$$

we obtain

$$\begin{aligned} |I_{\text{top}}| &\ll_{\eta, \varepsilon} \frac{x^c}{T} \sup_{\sigma_0(T) \leq \sigma \leq c} |F(\sigma + iT, y)| \\ &\ll_{\eta, \varepsilon} x^c T^{B-1}. \end{aligned}$$

The same computation on  $\mathcal{C}_{\text{bot}}$ , with  $s = \sigma - iT$ , gives

$$|I_{\text{bot}}| \ll_{\eta, \varepsilon} x^c T^{B-1}.$$

This proves (4.33). □

**Lemma 4.7.** For every  $T \geq 2$ ,

$$|I_{\text{left}}| \ll_{\eta, \varepsilon} x^{1-\lambda_T} T^B, \quad (4.34)$$

uniformly in  $|y| \leq \eta$ .

*Proof.* The contour  $\mathcal{C}_{\text{left}} = \mathcal{C}_{\text{left}}^- \cup \mathcal{C}_{\text{left}}^+$  is estimated by parameterizing the two boundary-value pieces as  $s = \sigma_0(T) + it$ , with  $-T \leq t < 0$  on  $\mathcal{C}_{\text{left}}^-$  and  $0 < t \leq T$  on  $\mathcal{C}_{\text{left}}^+$ . Using  $\sigma_0(T) = 1 - \lambda_T$ , we first write

$$\begin{aligned} |I_{\text{left}}| &= \left| \frac{1}{2\pi i} \int_{\mathcal{C}_{\text{left}}} F(s, y) \frac{x^s}{s} ds \right| \\ &\leq \frac{1}{2\pi} \left( \int_{-T}^{0^-} \left| F(\sigma_0(T) + it, y) \frac{x^{\sigma_0(T)+it}}{\sigma_0(T) + it} \right| dt + \int_{0^+}^T \left| F(\sigma_0(T) + it, y) \frac{x^{\sigma_0(T)+it}}{\sigma_0(T) + it} \right| dt \right) \\ &= \frac{x^{1-\lambda_T}}{2\pi} \left( \int_{-T}^{0^-} \left| \frac{F(\sigma_0(T) + it, y)}{\sigma_0(T) + it} \right| dt + \int_{0^+}^T \left| \frac{F(\sigma_0(T) + it, y)}{\sigma_0(T) + it} \right| dt \right). \end{aligned}$$

Since  $\sigma_0(T) > 1 - \delta' > 1/2$ , we have  $|\sigma_0(T) + it| \gg 1$  for  $|t| \leq 2$  and  $|\sigma_0(T) + it| \gg |t|$  for  $2 \leq |t| \leq T$ . Hence

$$|I_{\text{left}}| \ll x^{1-\lambda_T} \left( \int_{|t| \leq 2} |F(\sigma_0(T) + it, y)| dt + \int_{2 \leq |t| \leq T} \frac{|F(\sigma_0(T) + it, y)|}{|t|} dt \right).$$

On  $|t| \leq 2$ , Lemma 4.5 gives

$$|F(\sigma_0(T) + it, y)| \ll_{\eta, \delta'} (\log(T+2))^{2+2\eta}.$$

On  $2 \leq |t| \leq T$ , the point  $\sigma_0(T) + it$  lies in  $\mathcal{D}$  as a boundary-value limit and satisfies  $1 - \delta' \leq \sigma_0(T) \leq 1 + \delta'$ , so Lemma 4.5 gives

$$|F(\sigma_0(T) + it, y)| \ll_{\eta, \varepsilon} |t|^B.$$

Therefore

$$\begin{aligned} |I_{\text{left}}| &\ll_{\eta, \delta', \varepsilon} x^{1-\lambda_T} \left( (\log(T+2))^{2+2\eta} + \int_{2 \leq |t| \leq T} |t|^{B-1} dt \right) \\ &\ll_{\eta, \delta', \varepsilon} x^{1-\lambda_T} \left( (\log(T+2))^{2+2\eta} + 2 \int_2^T t^{B-1} dt \right). \end{aligned}$$

Since  $B > 0$ ,

$$2 \int_2^T t^{B-1} dt = \frac{2}{B} (T^B - 2^B) \ll_B T^B.$$

Also  $(\log(T+2))^{2+2\eta} \ll_{\eta, B} T^B$  for  $T \geq 2$ . Combining these estimates gives

$$|I_{\text{left}}| \ll_{\eta, \delta', \varepsilon} x^{1-\lambda_T} T^B.$$

This proves (4.34). □

### Choice of $T$ and the combined error

We choose

$$T = T(x) := \exp(\sqrt{\log x}), \tag{4.35}$$

the standard Selberg–Delange truncation height adapted to the de la Vallée Poussin zero-free region. With this choice,  $\log T = \sqrt{\log x}$ , and  $\lambda_T = c_0/(2\sqrt{\log x} + O(1))$ , so  $\lambda_T < \delta'$  holds automatically. Moreover,  $c = 1 + 1/\log x \leq 1 + \delta'$  for all  $x$  sufficiently large, so the abscissa  $c$  lies inside the strip used in the growth estimates. We have

$$x^{-\lambda_T} = \exp(-\lambda_T \log x) = \exp\left(-\frac{c_0}{2} \sqrt{\log x} \cdot (1 + o(1))\right). \tag{4.36}$$

**Lemma 4.8.** For each fixed integer  $K \geq 1$ , there exists a constant  $C_K > 0$  such that, for all sufficiently large  $X$ , if

$$C_K X^{1-1/(2K)} \leq H \leq X,$$

then

$$\sum_{X < n \leq X+H} d_K(n) \ll_K H(\log X)^{K-1}.$$

Here  $d_K(n)$  is defined by

$$\zeta(s)^K = \sum_{n \geq 1} d_K(n) n^{-s} \quad (\operatorname{Re}(s) > 1),$$

or equivalently as the number of ordered  $K$ -tuples of positive integers whose product is  $n$ .

*Proof.* We prove the assertion by induction on  $K$ . For  $K = 1$ , we have  $d_1(n) = 1$ , so the estimate is immediate.

Assume  $K \geq 2$  and that the result is known for  $K - 1$ . Put

$$D_K(X, H) := \sum_{X < n \leq X+H} d_K(n).$$

This counts ordered  $K$ -tuples  $(m_1, \dots, m_K)$  satisfying

$$X < m_1 \cdots m_K \leq X + H.$$

Since  $H \leq X$ , the product is at most  $2X$ , so at least one factor is at most  $(2X)^{1/K}$ . By the union bound over the  $K$  possible positions of such a factor,

$$D_K(X, H) \leq K \sum_{m \leq (2X)^{1/K}} \sum_{X/m < r \leq (X+H)/m} d_{K-1}(r).$$

Let  $Y = X/m$  and  $L = H/m$ . Choose  $C_K$  large enough, in terms of  $K$  and  $C_{K-1}$ , so that whenever  $H \geq C_K X^{1-1/(2K)}$  and  $m \leq (2X)^{1/K}$ , we have

$$L \geq C_{K-1} Y^{1-1/(2(K-1))}.$$

Also  $L \leq Y$ , since  $H \leq X$ . Hence the induction hypothesis applies to the inner sum and gives

$$\sum_{X/m < r \leq (X+H)/m} d_{K-1}(r) \ll_K \frac{H}{m} (\log X)^{K-2}.$$

Therefore

$$D_K(X, H) \ll_K H (\log X)^{K-2} \sum_{m \leq (2X)^{1/K}} \frac{1}{m} \ll_K H (\log X)^{K-1}.$$

□

**Lemma 4.9** (Perron truncation error). Let  $T = T(x) = \exp(\sqrt{\log x})$  and  $c = 1 + 1/\log x$ . For  $x \geq 2$  with  $x \notin \mathbb{Z}$  and  $|y| \leq \eta$ , there exists a fixed integer  $K = K(\eta)$  such that

$$E_{\text{Perron}}(x, T, y) \ll_{\eta} \frac{x}{T} (\log x)^K. \quad (4.37)$$

*Proof.* Set  $M = 1 + \eta$  and choose an integer  $K = K(\eta)$  with  $K \geq 2M$ . We first show that

$$d(n)M^{\omega(n)} \leq d_K(n) \quad (n \geq 1).$$

For a prime power  $p^a$  with  $a \geq 1$ ,

$$d(p^a)M^{\omega(p^a)} = (a+1)M, \quad d_K(p^a) = \binom{a+K-1}{K-1}.$$

The quantity

$$\frac{1}{a+1} \binom{a+K-1}{K-1}$$

is increasing in  $a$  for  $K \geq 3$ , and at  $a = 1$  it equals  $K/2$ . Thus  $(a+1)M \leq d_K(p^a)$  for every  $a \geq 1$ . Extending multiplicatively gives the claimed coefficient domination.

Using (4.6), we obtain

$$E_{\text{Perron}}(x, T, y) \ll_{\eta} x^c \sum_{n \geq 1} \frac{d_K(n)}{n^c} \frac{1}{1 + T \left| \log \frac{x}{n} \right|}.$$

We split the sum into the ranges  $n \leq x/2$ ,  $x/2 < n < 2x$ , and  $n \geq 2x$ .

If  $n \leq x/2$  or  $n \geq 2x$ , then  $|\log(x/n)| \geq \log 2$ , and hence the weight is  $\ll 1/T$ . The contribution of these two far ranges is therefore

$$\ll \frac{x^c}{T} \sum_{n \geq 1} \frac{d_K(n)}{n^c} = \frac{x^c}{T} \zeta(c)^K.$$

Since  $c = 1 + 1/\log x$ , we have  $x^c \asymp x$  and  $\zeta(c) \ll \log x$ , so the far contribution is

$$\ll_K \frac{x}{T} (\log x)^K.$$

It remains to estimate the central range  $x/2 < n < 2x$ . There  $x^c n^{-c} = (x/n)^c \ll 1$ , so it is enough to bound

$$\sum_{x/2 < n < 2x} d_K(n) \frac{1}{1 + T \left| \log \frac{x}{n} \right|}.$$

Put  $H = x/T$ . Cover the central range by the right intervals

$$I_r^+ = (x + rH, x + (r+1)H]$$

and the left intervals

$$I_r^- = [x - (r+1)H, x - rH),$$

with  $r = 0, 1, 2, \dots$  and only  $O(T)$  intervals needed. For  $n \in I_r^{\pm}$ ,

$$\frac{1}{1 + T \left| \log \frac{x}{n} \right|} \ll \frac{1}{1 + r}.$$

For  $r = 0$  this is trivial. For  $r \geq 1$ , since  $x/2 < n < 2x$ , we have

$$\left| \log \frac{x}{n} \right| \asymp \frac{|n - x|}{x} \asymp \frac{rH}{x} = \frac{r}{T}.$$

Because  $T = \exp(\sqrt{\log x})$ , the length  $H = x \exp(-\sqrt{\log x})$  is larger than  $C_K x^{1-1/(2K)}$  for all sufficiently large  $x$ . Each interval lies in  $x/2 < n < 2x$ , so Lemma 4.8 applies with a starting point comparable to  $x$ , and gives

$$\sum_{n \in I_r^{\pm}} d_K(n) \ll_K H (\log x)^{K-1}.$$

Therefore the central contribution is

$$\ll_K H (\log x)^{K-1} \sum_{r \ll T} \frac{1}{1 + r} \ll_K \frac{x}{T} (\log x)^{K-1} \log T.$$

Since  $\log T = \sqrt{\log x} \leq \log x$  for sufficiently large  $x$ , this is

$$\ll_K \frac{x}{T} (\log x)^K.$$

Combining the far and central estimates proves (4.37).  $\square$

**Proposition 4.10.** With  $T = T(x) = \exp(\sqrt{\log x})$ , there exists a constant  $c_1 > 0$  (depending on  $\eta, \varepsilon$  but not on  $x$ ) such that

$$E_{\text{Perron}}(x, T, y) + |I_{\text{top}}| + |I_{\text{bot}}| + |I_{\text{left}}| \ll_{\eta, \varepsilon} x \exp(-c_1 \sqrt{\log x}), \quad (4.38)$$

uniformly in  $|y| \leq \eta$ , for all sufficiently large  $x \notin \mathbb{Z}$ .

*Proof.* We bound each term separately and take the worst.

*Perron error.* By Lemma 4.9 with  $T = \exp(\sqrt{\log x})$ ,

$$E_{\text{Perron}} \ll_{\eta} x \exp(-\sqrt{\log x})(\log x)^K.$$

Since  $(\log x)^K = \exp(K \log \log x)$  and, for sufficiently large  $x$ ,

$$K \log \log x \leq \frac{1}{2} \sqrt{\log x},$$

we have

$$E_{\text{Perron}} \ll_{\eta} x \exp\left(-\frac{1}{2} \sqrt{\log x}\right).$$

*Horizontal pieces.* With  $c = 1 + 1/\log x$  so that  $x^c = e \cdot x$ , by Lemma 4.6,

$$|I_{\text{top}}| + |I_{\text{bot}}| \ll_{\eta, \varepsilon} x \cdot T^{B-1} = x \cdot \exp((B-1)\sqrt{\log x}).$$

Since  $B < 1$ , this is  $\ll x \exp(-(1-B)\sqrt{\log x})$ .

*Vertical piece.* By Lemma 4.7 and (4.36),

$$|I_{\text{left}}| \ll_{\eta, \varepsilon} x \cdot x^{-\lambda_T} \cdot T^B = x \cdot \exp\left(\left(-\frac{c_0}{2} + B + o(1)\right)\sqrt{\log x}\right).$$

By the choice (4.13) of  $\delta'$  together with (4.32), we have  $B < c_0/2$ . Hence  $|I_{\text{left}}| \ll x \exp(-(c_0/2 - B)\sqrt{\log x})$ .

Set

$$c_1 := \frac{1}{2} \min\left(\frac{1}{2}, 1 - B, \frac{c_0}{2} - B\right) > 0.$$

For all sufficiently large  $x$ , the preceding three estimates are bounded by the right-hand side of (4.38).  $\square$

The right-hand side of (4.38) is exponentially smaller than the main contribution to  $A(x, y)$ . The local Hankel contribution  $I_H$ , analyzed in Section 5, contains the main term from which the asymptotic formula for each fixed  $T_j(x)$  will later be extracted.

## 5 Local Hankel analysis

The purpose of this section is to analyze the local Hankel contribution  $I_H$  around the singular point  $s = 1$ . Recall that

$$F(s, y) = \zeta(s)^{2+2y} H(s, y).$$

The singular behavior is carried by the power of  $\zeta(s)$ . To isolate it, we factor the remaining part as  $G(s, y) := ((s-1)\zeta(s))^\alpha H(s, y)$ , which extends holomorphically to  $s = 1$ .

## 5.1 The local regular factor

Define

$$q(s) := (s-1)\zeta(s), \quad q(1) := 1. \quad (5.1)$$

The pole of  $\zeta(s)$  at  $s = 1$  is simple with residue 1, so  $q$  is holomorphic near  $s = 1$  after filling in the removable singularity. After shrinking the neighborhood of  $s = 1$  if necessary, we may assume that  $q(s) \neq 0$  there. We then choose a single branch of  $\log q(s)$  in this neighborhood, normalized by

$$\log q(1) = 0.$$

For  $s$  in this neighborhood and  $y$  near 0, define

$$G(s, y) := q(s)^{2+2y} H(s, y), \quad q(s)^{2+2y} := \exp((2+2y) \log q(s)). \quad (5.2)$$

On the corresponding punctured slit neighborhood of  $s = 1$ , the branch choices from Section 4 give

$$F(s, y) = G(s, y)(s-1)^{-2-2y}. \quad (5.3)$$

Here the branch of  $(s-1)^{-2-2y}$  is the local Hankel branch induced by the contour and branch conventions already fixed in Section 4.3.

**Lemma 5.1.** There exist  $r \in (0, \delta)$  and  $C_{\eta, \delta} > 0$  such that  $G(s, y)$  is holomorphic in  $(s, y)$  for  $|s-1| < r$  and  $|y| < \eta$ , and

$$|G(s, y)| \leq C_{\eta, \delta}$$

uniformly for  $|s-1| \leq r$  and  $|y| \leq \eta$ . Moreover, on the punctured slit neighborhood

$$0 < |s-1| < r, \quad s \notin \mathcal{S},$$

we have the local factorization

$$F(s, y) = G(s, y)(s-1)^{-2-2y}.$$

*Proof.* Apply Proposition 3.4 with this fixed  $\delta$ . It gives that the function  $H(s, y)$  is holomorphic in  $(s, y)$  on the interior of  $\mathcal{R}_{\delta, \eta}$  and uniformly bounded on  $\mathcal{R}_{\delta, \eta}$ . Choose  $r \in (0, \delta)$ . Then  $|s-1| \leq r$  implies  $\operatorname{Re}(s) \geq 1-r > 1-\delta$ , so

$$H(s, y) \ll_{\eta, \delta} 1 \quad (|s-1| \leq r, |y| \leq \eta).$$

The function  $q(s) = (s-1)\zeta(s)$  is holomorphic at  $s = 1$  after setting  $q(1) = 1$ . Since  $q(1) = 1 \neq 0$ , we shrink  $r$ , preserving  $r \in (0, \delta)$ , so that  $q$  has no zeros on  $|s-1| \leq r$ . Then a holomorphic branch of  $\log q(s)$  exists on  $|s-1| < r$ , normalized by  $\log q(1) = 0$ . Since this branch is holomorphic on a neighborhood of the closed disk  $|s-1| \leq r$ , it is bounded there. Therefore

$$q(s)^{2+2y} = \exp((2+2y) \log q(s))$$

is uniformly bounded for  $|s-1| \leq r$  and  $|y| \leq \eta$ . Combining this local bound with the bound for  $H(s, y)$  gives  $|G(s, y)| \leq C_{\eta, \delta}$ .

Finally, on the punctured slit neighborhood,

$$\zeta(s)^{2+2y} = q(s)^{2+2y}(s-1)^{-2-2y},$$

with the branch of  $(s-1)^{-2-2y}$  inherited from the local Hankel contour. Combining this identity with  $F(s, y) = \zeta(s)^{2+2y} H(s, y)$  proves the asserted factorization.  $\square$

**Lemma 5.2.** Let  $r \in (0, \delta)$  be chosen as in Lemma 5.1. Then

$$G(1, y) = H(1, y)$$

for  $|y| \leq \eta$ , and

$$G(1, 0) = 1.$$

Moreover,

$$G(s, y) = G(1, y) + O_{\eta, \delta}(|s - 1|)$$

uniformly for  $|s - 1| \leq r/2$  and  $|y| \leq \eta$ .

*Proof.* Since  $q(1) = 1$  and the branch is normalized by  $\log q(1) = 0$ , we have

$$G(1, y) = q(1)^{2+2y} H(1, y) = H(1, y).$$

By Lemma 3.5,  $H(1, 0) = 1$ . Therefore  $G(1, 0) = H(1, 0) = 1$ .

For the first-order estimate, use Lemma 5.1. Since  $G$  is holomorphic in  $s$  on  $|s - 1| < r$  and uniformly bounded for  $|s - 1| \leq r$ ,  $|y| \leq \eta$ , Cauchy's estimate in the  $s$ -variable gives

$$\frac{\partial G}{\partial s}(s, y) \ll_{\eta, \delta} 1$$

uniformly for  $|s - 1| \leq r/2$  and  $|y| \leq \eta$ . Therefore

$$G(s, y) - G(1, y) = \int_1^s \frac{\partial G}{\partial s}(w, y) dw = O_{\eta, \delta}(|s - 1|),$$

uniformly in the stated range. □

**Corollary 5.3.** Let

$$B_1 = \gamma + \sum_p \left( \log \left( 1 - \frac{1}{p} \right) + \frac{1}{p} \right)$$

be the Meissel–Mertens constant [6], and let

$$P(2) = \sum_p \frac{1}{p^2}$$

be the value of the prime zeta function at 2. Then

$$\partial_y H(1, 0) = 2(B_1 - \gamma) - P(2).$$

Consequently,

$$\partial_y H(1, 0) - 2(1 - \gamma) = 2(B_1 - 1) - P(2).$$

*Proof.* For  $u = p^{-s}$ , the local factor in the Euler product for  $H$  is

$$H_p(s, y) = (1 - u)^{2+2y} (1 + (1 + y) ((1 - u)^{-2} - 1)).$$

For every prime  $p$ , we have  $H_p(1, 0) = 1$ , and hence  $H(1, 0) = 1$ . Logarithmic differentiation of the Euler product gives

$$\partial_y H(1, 0) = \sum_p \partial_y \log H_p(1, 0).$$

Indeed, the summand computed below is  $O(p^{-2})$ , so the series converges absolutely and the logarithmic differentiation is justified.

We compute

$$\partial_y \log H_p(s, y)|_{y=0} = 2 \log(1 - u) + \frac{(1 - u)^{-2} - 1}{(1 - u)^{-2}}.$$

Substituting  $u = 1/p$  gives

$$\partial_y \log H_p(1, 0) = 2 \log \left(1 - \frac{1}{p}\right) + \frac{2}{p} - \frac{1}{p^2}.$$

Therefore

$$\partial_y H(1, 0) = \sum_p \left(2 \log \left(1 - \frac{1}{p}\right) + \frac{2}{p} - \frac{1}{p^2}\right).$$

Using the defining relation

$$\sum_p \left(\log \left(1 - \frac{1}{p}\right) + \frac{1}{p}\right) = B_1 - \gamma,$$

we obtain

$$\partial_y H(1, 0) = 2(B_1 - \gamma) - P(2).$$

Subtracting  $2(1 - \gamma)$  gives the final displayed identity.  $\square$

## 5.2 Evaluation of the local Hankel contribution

**Proposition 5.4.** Put

$$T = \exp(\sqrt{\log x}), \quad \rho := \frac{1}{\log x}.$$

For all sufficiently large  $x$ , uniformly for  $|y| \leq \eta$ , we have

$$I_H = xH(1, y) \frac{(\log x)^{1+2y}}{\Gamma(2+2y)} + O_\eta(x(\log x)^{2\eta}).$$

*Proof.* We apply Lemmas 5.1 and 5.2, the constants below depend only on  $\eta$ . Let  $r$  be the local radius supplied by these lemmas. With  $T = \exp(\sqrt{\log x})$ , we have

$$\lambda_T = \frac{c_0}{2 \log(T+2)} \asymp \frac{1}{\sqrt{\log x}}.$$

Thus  $\rho = 1/\log x < \lambda_T$  and  $\rho < r/2$  for all sufficiently large  $x$ .

By the definition of  $I_H$  in (4.26), together with the local factorization recorded in Lemma 5.1,

$$\begin{aligned} I_H &:= \frac{1}{2\pi i} \int_{\mathcal{H}_{\text{loc}}} F(s, y) \frac{x^s}{s} ds \\ &= \frac{1}{2\pi i} \int_{\mathcal{H}_{\text{loc}}} G(s, y) (s-1)^{-\alpha} \frac{x^s}{s} ds \\ &= \frac{x}{2\pi i} \int_{\mathcal{H}_z} \frac{G(1+z, y)}{1+z} z^{-\alpha} e^{z \log x} dz, \end{aligned}$$

where the last equality follows from the change of variables  $z = s - 1$ . Here  $\mathcal{H}_z$  is the image of  $\mathcal{H}_{\text{loc}}$  under this change of variables. Thus  $\mathcal{H}_z$  is the truncated Hankel contour whose arms run along the



negative real axis from  $-\lambda_T$  to  $-\rho$ , with the circle  $|z| = \rho$  oriented counterclockwise through the right side of the origin.

For  $z$  on  $\mathcal{H}_z$ , Lemma 5.2 gives

$$G(1+z, y) = G(1, y) + O_\eta(|z|).$$

Also, since  $|z| \leq \lambda_T < 1/2$  for large  $x$ ,

$$\frac{1}{1+z} = 1 + O(|z|).$$

Using the uniform boundedness of  $G(1, y)$ , we obtain

$$\frac{G(1+z, y)}{1+z} = G(1, y) + E(z, y), \quad E(z, y) = O_\eta(|z|), \quad (5.4)$$

uniformly for  $|y| \leq \eta$  and  $z \in \mathcal{H}_z$ .

Accordingly, write

$$I_H = M_H + R_H,$$

where

$$M_H := \frac{xG(1, y)}{2\pi i} \int_{\mathcal{H}_z} z^{-\alpha} e^{z \log x} dz$$

and

$$R_H := \frac{x}{2\pi i} \int_{\mathcal{H}_z} E(z, y) z^{-\alpha} e^{z \log x} dz.$$

We first evaluate  $M_H$ . Put

$$U := \lambda_T \log x.$$

Under the change of variables  $w = z \log x$ , the contour  $\mathcal{H}_z$  is mapped to a truncated Hankel contour  $\mathcal{H}_U$  in the  $w$ -plane, whose arms run from  $-U$  to  $-1$  and whose circular part is  $|w| = 1$ . Then

$$M_H = xG(1, y)(\log x)^{\alpha-1} \frac{1}{2\pi i} \int_{\mathcal{H}_U} e^w w^{-\alpha} dw.$$

Let  $\mathcal{H}_\infty$  be the corresponding infinite Hankel contour. By the standard Hankel representation of  $1/\Gamma(\alpha)$  [3, p.139],

$$\frac{1}{2\pi i} \int_{\mathcal{H}_\infty} e^w w^{-\alpha} dw = \frac{1}{\Gamma(\alpha)}.$$

The difference between  $\mathcal{H}_\infty$  and  $\mathcal{H}_U$  consists of the upper and lower tails over  $(-\infty, -U]$ . On these tails write  $w = -u \pm i0$ , with  $u \geq U$ . Then  $|e^w| = e^{-u}$ . Also, since  $|y| \leq \eta$  and  $\alpha = 2 + 2y$ , the branch factors  $e^{\pm i\pi\alpha}$  are bounded by a constant depending only on  $\eta$ , and hence

$$|w^{-\alpha}| \ll_\eta u^{-\operatorname{Re}(\alpha)}.$$

After taking absolute values, the two tails and the factor  $1/(2\pi i)$  are absorbed into the  $O_\eta$ -constant, giving

$$\left| \frac{1}{2\pi i} \int_{\mathcal{H}_U} e^w w^{-\alpha} dw - \frac{1}{\Gamma(\alpha)} \right| \ll_\eta \int_U^\infty e^{-u} u^{-\operatorname{Re}(\alpha)} du.$$

Since

$$\operatorname{Re}(\alpha) = 2 + 2\operatorname{Re}(y) \geq 2 - 2\eta > 1$$

and  $U \geq 1$  for sufficiently large  $x$ , we have

$$\int_U^\infty e^{-u} u^{-\operatorname{Re}(\alpha)} du \leq \int_U^\infty e^{-u} du = e^{-U}.$$

Therefore

$$\frac{1}{2\pi i} \int_{\mathcal{H}_U} e^w w^{-\alpha} dw = \frac{1}{\Gamma(\alpha)} + O_\eta(e^{-U}).$$

It follows that

$$M_H = xG(1, y) \frac{(\log x)^{\alpha-1}}{\Gamma(\alpha)} + O_\eta \left( x(\log x)^{\operatorname{Re}(\alpha)-1} e^{-U} \right).$$

Since  $U = \lambda_T \log x \asymp \sqrt{\log x}$ , there is a constant  $c > 0$  such that  $e^{-U} \ll e^{-c\sqrt{\log x}}$ . Also  $\operatorname{Re}(\alpha) - 1 \leq 1 + 2\eta$ . Hence the tail contribution is

$$\ll_\eta x(\log x)^{1+2\eta} e^{-c\sqrt{\log x}} = O_\eta \left( x(\log x)^{2\eta} \right),$$

because  $(\log x)e^{-c\sqrt{\log x}} = O(1)$ .

It remains to estimate  $R_H$ . From (5.4),

$$|R_H| \ll_\eta x \int_{\mathcal{H}_z} |z| |z^{-\alpha}| e^{\operatorname{Re}(z) \log x} |dz|.$$

On the Hankel contour,  $|\operatorname{Arg} z| \leq \pi$  and  $|\Im \alpha| \leq 2\eta$ , so

$$|z^{-\alpha}| \ll_\eta |z|^{-\operatorname{Re}(\alpha)}.$$

Thus

$$|R_H| \ll_\eta x \int_{\mathcal{H}_z} |z|^{1-\operatorname{Re}(\alpha)} e^{\operatorname{Re}(z) \log x} |dz|.$$

Changing variables again by  $w = z \log x$  gives

$$|R_H| \ll_\eta x(\log x)^{\operatorname{Re}(\alpha)-2} \int_{\mathcal{H}_U} |w|^{1-\operatorname{Re}(\alpha)} e^{\operatorname{Re} w} |dw|.$$

The last integral is  $O_\eta(1)$ : this is immediate on  $|w| = 1$ , and on the arms it is bounded by

$$\int_1^U u^{1-\operatorname{Re}(\alpha)} e^{-u} du \leq \int_1^\infty u^{1-\operatorname{Re}(\alpha)} e^{-u} du \ll_\eta 1.$$

Consequently,

$$R_H = O_\eta \left( x(\log x)^{\operatorname{Re}(\alpha)-2} \right).$$

Since  $\alpha = 2 + 2y$ , we have

$$\operatorname{Re}(\alpha) - 2 = 2 \operatorname{Re}(y) \leq 2\eta,$$

and hence

$$R_H = O_\eta(x(\log x)^{2\eta}).$$

Combining the estimates for  $M_H$  and  $R_H$ , and using Lemma 5.2 to replace  $G(1, y)$  by  $H(1, y)$ , we obtain

$$I_H = xH(1, y) \frac{(\log x)^{1+2y}}{\Gamma(2+2y)} + O_\eta(x(\log x)^{2\eta}),$$

uniformly for  $|y| \leq \eta$ . □

**Proposition 5.5** (Generating-function asymptotic). With the parameters chosen as in Sections 4 and 5, and with the fixed contour parameters suppressed in the implied constant, uniformly for  $|y| \leq \eta$ , as  $x \rightarrow \infty$ ,

$$A(x, y) = xH(1, y) \frac{(\log x)^{1+2y}}{\Gamma(2+2y)} + O_\eta(x(\log x)^{2\eta}).$$

*Proof.* For non-integral  $x$ , the contour decomposition (4.24) gives

$$A(x, y) = I_H + I_{\text{top}} + I_{\text{bot}} + I_{\text{left}} + E_{\text{Perron}}(x, T, y).$$

By Proposition 4.10, for some  $c_1 > 0$ ,

$$E_{\text{Perron}}(x, T, y) + |I_{\text{top}}| + |I_{\text{bot}}| + |I_{\text{left}}| \ll_\eta x \exp(-c_1 \sqrt{\log x}),$$

uniformly for  $|y| \leq \eta$ . Proposition 5.4 gives

$$I_H = xH(1, y) \frac{(\log x)^{1+2y}}{\Gamma(2+2y)} + O_\eta(x(\log x)^{2\eta}).$$

The exponentially small nonlocal and Perron error is absorbed into  $O_\eta(x(\log x)^{2\eta})$ . Therefore the asserted asymptotic holds for non-integral  $x$ .

For integral  $x$ , the replacement by  $x + 1/2$  described in Section 4.1 gives the same estimate, since  $A(x, y) = A(x + 1/2, y)$  and replacing  $x + 1/2$  by  $x$  changes the displayed main term by an amount absorbed into the same error term.  $\square$

## 6 Extraction of $T_j(x)$ and proof of the main theorem

*Proof of Theorem 1.3.* By Proposition 5.5, uniformly for  $|y| \leq \eta$ ,

$$A(x, y) = xH(1, y) \frac{(\log x)^{1+2y}}{\Gamma(2+2y)} + O_\eta(x(\log x)^{2\eta}).$$

Define

$$K(y) := \frac{H(1, y)}{\Gamma(2+2y)}.$$

Then  $K$  is holomorphic in a neighborhood of 0, and

$$A(x, y) = x \log x K(y) e^{2y \log \log x} + O_\eta(x(\log x)^{2\eta}).$$

The generating relation

$$A(x, y) = \sum_{j \geq 0} T_j(x) y^j$$

gives the coefficient identity below. Here  $[y^j]$  denotes the coefficient of  $y^j$  in the power series expansion:

$$T_j(x) = [y^j] A(x, y).$$

Write

$$K(y) = \sum_{r \geq 0} k_r y^r, \quad k_r = \frac{K^{(r)}(0)}{r!},$$

and

$$e^{2y \log \log x} = \sum_{m \geq 0} \frac{(2 \log \log x)^m}{m!} y^m.$$

For fixed  $j$ ,

$$[y^j] \left( K(y) e^{2y \log \log x} \right) = \sum_{r=0}^j k_r \frac{(2 \log \log x)^{j-r}}{(j-r)!}.$$

Therefore

$$[y^j] \left( K(y) e^{2y \log \log x} \right) = k_0 \frac{(2 \log \log x)^j}{j!} + k_1 \frac{(2 \log \log x)^{j-1}}{(j-1)!} + O_j((\log \log x)^{j-2}).$$

It remains to compute  $k_0$  and  $k_1$ . By Lemma 3.5,  $H(1, 0) = 1$ . Also  $\Gamma(2) = 1$ . Hence

$$k_0 = K(0) = 1.$$

Next,

$$k_1 = K'(0) = \partial_y H(1, 0) + H(1, 0) \frac{d}{dy} \frac{1}{\Gamma(2 + 2y)} \Big|_{y=0}.$$

Here  $\psi = \Gamma'/\Gamma$  denotes the digamma function. Since

$$\frac{d}{dy} \frac{1}{\Gamma(2 + 2y)} \Big|_{y=0} = -2\psi(2) = -2(1 - \gamma),$$

we get

$$k_1 = \partial_y H(1, 0) - 2(1 - \gamma).$$

By Corollary 5.3, this is

$$k_1 = 2(B_1 - 1) - P(2).$$

We next extract  $j$ -th coefficients of the error term. Choose a fixed radius  $\rho_y$  with  $0 < \rho_y < \eta$ . Since the error in Proposition 5.5 is uniform for  $|y| \leq \eta$ , Cauchy's coefficient estimate on  $|y| = \rho_y$  gives, for each fixed  $j$ ,

$$[y^j] O_\eta(x(\log x)^{2\eta}) = O_{\eta, j, \rho_y}(x(\log x)^{2\eta}).$$

Because  $0 < \eta < 1/2$ , for fixed  $j \geq 2$ ,

$$x(\log x)^{2\eta} = O_j(x \log x (\log \log x)^{j-2}).$$

Thus this error is absorbed into the final error term.

Combining the preceding estimates gives, for each fixed  $j \geq 2$ ,

$$\begin{aligned} T_j(x) &= \frac{2^j}{j!} x \log x (\log \log x)^j \\ &\quad + \frac{2^{j-1}}{(j-1)!} (2(B_1 - 1) - P(2)) x \log x (\log \log x)^{j-1} \\ &\quad + O_j(x \log x (\log \log x)^{j-2}). \end{aligned}$$

In particular,

$$T_j(x) \sim \frac{2^j}{j!} x \log x (\log \log x)^j.$$

□

## 7 Toward $D(x)$ : outlook

The identity (1.6) expresses  $D(x)$  exactly in terms of the prime-power contribution and the alternating finite sum of the quantities  $T_j(x)$ . Theorem 1.3, however, is a fixed- $j$  asymptotic statement. Since the range of relevant  $j$  grows with  $x$ , these fixed- $j$  asymptotics cannot be inserted term by term into (1.6) to obtain an asymptotic formula for  $D(x)$ .

Equivalently, if

$$A(x, y) = \sum_{j \geq 0} T_j(x) y^j,$$

then the alternating part of (1.6) is determined by values of  $A(x, y)$  and its  $y$ -derivative at  $y = -1$ . The analytic work in this paper is carried out for  $y$  in a small neighborhood of 0. Extending the method toward  $D(x)$  would therefore require estimates uniform in  $j$ , or a direct analysis of the generating function near  $y = -1$ .

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